

NIELIT Virtual Academy

COURSE PROSPECTUS

Name of the Course: O level - *IT TOOLS & Network Basics*

Course Code: *ITB41*

Mode of Conduction: *Online[Self-Paced]*

Duration: *120 Hours*

Objective of the Course

The goal of this course is to present overview of IT tools used in day to day use of Computers. The Course has been designed to provide knowledge on various hardware and software components of computer, operating system, various packages used for different applications, and various issues related to IT and application of IT.

At the end of the course the students will be able to:-

- ✓ Acquire the foundation level knowledge required to understand computer and its operations.
- ✓ Understand the hardware and software components of the computer.
- ✓ Understand the basic concept of operating system and get knowledge about various different operating systems.
- ✓ Understand to use the packages of word processing, spread sheet and presentation in detail.
- ✓ Understand the issues related to IT and IT applications.

Outcome of the Course:

An IT Tools and Network Basics course aims to equip individuals with fundamental computer and networking knowledge. Upon completion, students typically acquire the following skills and understanding:

- ✓ Proficiency in basic software applications like Word processing, spreadsheets, presentations, email, and web browsing.
- ✓ Understanding of software concepts and their functions.
- ✓ Understanding network components (hardware and software), network topologies, and communication protocols.
- ✓ Internet concepts: How the internet works, basic internet services (email, web browsing, etc.), and internet security.
- ✓ Digital literacy: Safe and effective use of computers and digital resources.
- ✓ Awareness of emerging trends and technologies in IT and networking, such as cloud computing, AI, Big Data Analysis IoT, Cyber Security etc.

Course Fee: Rs: 590/-

Eligibility:

12th

Or ITI Certificate (Two Years) after class 10

Or ITI Certificate (One Years) after class 10 with one year of experience post qualification. Or Successful completion of the second year of a Government recognized

polytechnic engineering diploma course after class 10, Training of 'O' Level course concurrently during the third year of the said 3 years Polytechnic engineering diploma course.

Methodology:

- ✓ Teaching Mode: Self-Pace
- ✓ Access from anywhere anytime
- ✓ Content Access through e-learning portal
- ✓ Doubt Clearing Session
- ✓ Practical Oriented

Registration Link: <http://nva.nielit.gov.in>

Contact Details:

- Course coordinator Name: Smt. Priyanka Srivastava
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Contents:

Module: M1-R5.1 IT Tools and Network Basics	
Computer Appreciation and Organization	
UNIT 1	Computer and Latest IT gadgets, Evolution of Computers & its applications, IT gadgets and their applications, Basics of Hardware and Software, Central Processing Unit, Input devices, Output devices, Computer Memory & storage, Application Software, Systems Software, Utility Software, Open source and Proprietary Software, Mobile Apps.
Introduction to Operating System	
UNIT 2	Operating System, Basics of Operating System, Operating Systems for Desktop and Laptop, Operating Systems for Mobile Phone and Tablets, User Interface for Desktop and Laptop, Task Bar, Icons & shortcuts, running an application, Operating System simple setting, using mouse and changing its properties, changing system date and time, changing display properties, Add or remove Program and its features, adding, removing & sharing Printers, File and Folder management, types of file extensions.
Word Processing	
UNIT 3	Word Processing Basics, Opening Word Processing Package, Title Bar, Menu Bar, Toolbars & Sidebar, Creating a New Document, Opening and Closing Documents, Opening Documents, Save and Save As, Closing Document, Using The Help, Page Setup, Page Layout, Borders, Watermark, Print Preview, Printing of Documents, PDF file and Saving a Document as PDF file, Text Creation and manipulation, Document Creation, Editing Text, Text Selection, Cut, Copy and Paste, Font, Color, Style and Size selection, Alignment of Text, Undo & Redo, AutoCorrect, Spelling & Grammar, Find and Replace. Formatting the Text, Creating and using user defined Styles, Paragraph Indentation, Bullets and Numbering, Change case, Header & Footer Table Manipulation, Insert & Draw Table, Changing cell width and height, Alignment of Text in cell, Delete / Insertion of Row, Column and Merging & Splitting of Cells, Border and Shading, Mail Merge, Table of Contents, Indexes, Adding Comments, Tracking

	changes, Macros
Spreadsheet	
UNIT 4	Elements of Spread Sheet, Creating of Spread Sheet, Concept of Cell Address [Row and Column] and selecting a Cell, Entering Data [text, number, date] in Cells, Page Setup, Printing of Sheet, Saving Spreadsheet, Opening and Closing, Manipulation of Cells & Sheet, Modifying / Editing Cell Content , Formatting Cell (Font, Alignment, Style), Cut, Copy, Paste & Paste Special, Changing Cell Height and Width, Inserting and Deleting Rows, Column, AutoFill Sorting & Filtering, Freezing panes ,Formulas, Functions and Charts, Using Formulas for Numbers (Addition, Subtraction, Multiplication & Division), AutoSum, Functions (Sum, Count, MAX, MIN, AVERAGE),Sort, Filter, Advanced Filter, Database Functions (DSUM, DMIN,DMAX, DCOUNT, DCOUNTA) What-if Analysis, Pivot table Charts (Bar, Column, Pie, Line), Data Validation.
Presentation	
UNIT 5	Creation of Presentation, Creating a Presentation Using a Template, Creating a Blank Presentation, Inserting & Editing Text on Slides, Inserting and Deleting Slides in a Presentation, Saving a Presentation, Manipulating Slides, Inserting Table , Adding Pictures, Inserting Other Objects, Resizing and Scaling an Object Creating & using Master Slide, Presentation of Slides , Choosing a Set Up for Presentation, Running a Slide Show Transition and Slide Timings, Automating a Slide Show, Providing Aesthetics to Slides & Printing, Enhancing Text Presentation, Working with Color and Line Style, Adding Movie and Sound, Adding Headers, Footers and Notes, Printing Slides and Handouts
Introduction to Internet and WWW	
UNIT 6	Basic of Computer Networks, Local Area Network (LAN), Wide Area Network (WAN), Network Topology Internet, Concept of Internet & WWW, Applications of Internet, Website Address and URL, Introduction to IP Address ISP and Role of ISP, Internet Protocol, Modes of Connecting Internet (HotSpot, Wifi, LAN Cable, BroadBand, USB Tethering), Identifying and uses of IP/MAC/IMEI of various devices Popular Web Browsers (Internet Explorer/Edge, Chrome, Mozilla Firefox, Opera etc.), Exploring the Internet , Surfing the web, Popular Search Engines, Searching on Internet, Downloading Web Pages, Printing Web Pages
E-mail, Social Networking and e-Governance Services	
UNIT 7	Structure of E-mail, Using E-mails, Opening Email account, Mailbox: Inbox and Outbox, Creating and Sending a new E-mailReplying to an E-mail message, Forwarding an E-mail message, Searching emails, Attaching files with email, Email Signature Social Networking & e-Commerce, Facebook, Twitter, Linkedin, Instagram, Instant Messaging (Whatsapp, Facebook Messenger, Telegram), Introduction to Blogs Basics of E-commerce, Netiquettes, Overview of e-Governance, Services like Railway Reservation, Passport, e-Hospital [ORS], Accessing e-Governance Services on Mobile Using “UMANG APP”, Digital Locker
Digital Financial Tools and Applications	

UNIT 8	Digital Financial Tools, Understanding OTP [One Time Password] and QR [Quick Response] Code, UPI [Unified Payment Interface], AEPS [Aadhaar Enabled Payment System], USSD [Unstructured Supplementary Service Data], Card [Credit / Debit], eWallet, PoS [Point of Sale] Internet Banking, National Electronic Fund Transfer (NEFT), Real Time Gross Settlement (RTGS), Immediate Payment Service (IMPS), Online Bill Payment
Overview of Futureskills and Cyber Security	
UNIT 9	Introduction to Internet of Things (IoT), Big Data Analytics, Cloud Computing, Virtual Reality, Artificial Intelligence, Social & Mobile Blockchain Technology, 3D Printing/ Additive Manufacturing, Robotics Process Automation Cyber Security, Need of Cyber Security, Securing PC, Securing Smart Phone

Examination & Certification

Category	Theory	Total
M1-R5.1: Information Technology Tools and Network Basics	01	100
Marks	100	100

After successful completion of the course, candidate will get an online certificate with the following Grading Scheme:

Marks Range	Grade	Certificate Type
85% and above	S	Graded
75-84%	A	Graded
65-74%	B	Graded
55-64%	C	Graded
50-54%	D	Graded
<50%	F	Participation
Attended the Course but not appeared in Examination	N	Participation